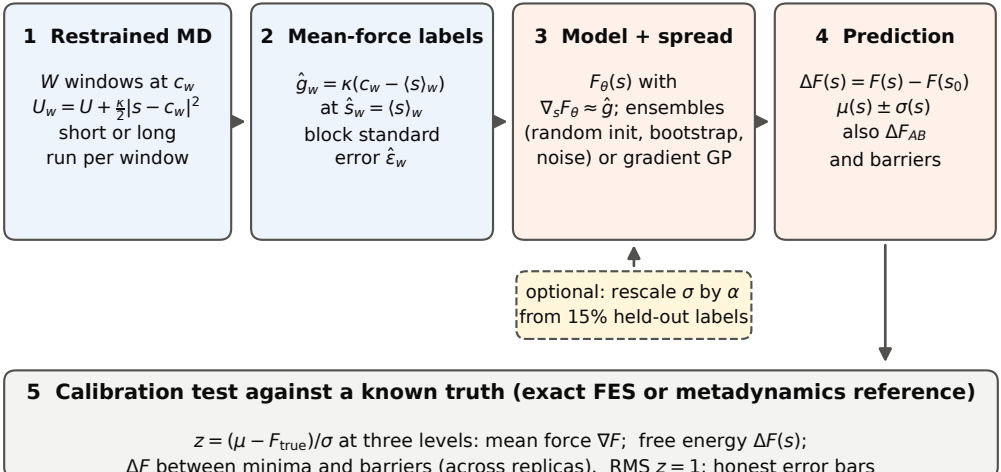
1 Restrained MD
W windows at c_w
$U_w = U + \frac{\kappa}{2}|s - c_w|^2$
short or long
run per window
2 Mean-force labels
$\hat{g}_w = \kappa(c_w - \langle s \rangle_w)$
at $\hat{s}_w = \langle s \rangle_w$
block standard
error $\hat{\varepsilon}_w$
3 Model + spread
$F_\theta(s)$ with
$\nabla_s F_\theta \approx \hat{g}$; ensembles
(random init, bootstrap,
noise) or gradient GP
4 Prediction
$\Delta F(s) = F(s) - F(s_0)$
$\mu(s) \pm \sigma(s)$
also ΔF_{AB}
and barriers
optional: rescale σ by α
from 15% held-out labels
5 Calibration test against a known truth (exact FES or metadynamics reference)
$z = (\mu - F_{\text{true}})/\sigma$ at three levels: mean force ∇F; free energy $\Delta F(s)$;
ΔF between minima and barriers (across replicas). RMS $z = 1$: honest error bars